

Report of libRosa Analysis

Candy Xie

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Link of notebook: [🔗 Librosa Assignment Candy Xie](#)

Track 1: "By My Side" by Tyson Yoshi (Cantopop/R&B)

Tempo Estimation

Librosa estimated the tempo at 165.44 BPM. The actual tempo of the track is approximately 82 BPM. This implied a classic octave error, in which the algorithm locks onto a rhythmic subdivision (likely eighth note pulses from the hi-hat pattern) rather than the perceived quarter note beat. Yet, by dividing the estimate by two yields 82.7 BPM, it matches the true tempo.

Onset Detection (63 onsets in 10 seconds, segment from 45s to 55s)

The onset detection identified 63 events across the 10-second segment, averaging roughly 6 onsets per second. This is a high count, but it reflects the density of the mix with layered drum machine hits, vocal syllables, and synthesizer attacks all register as separate onset events. Based on the waveform plot, the red onset markers generally align with visible amplitude spikes, particularly where drum hits and vocal consonants occur. The algorithm performed well on rhythmically active moments with clear transients. Some of the detected onsets likely correspond to subtle hi-hat or percussion hits that are individually hard to verify by ear but are present in the waveform. Overall, the onset detection was largely accurate for this track, which is expected given the steady, well-defined rhythmic structure of pop or R&B production.

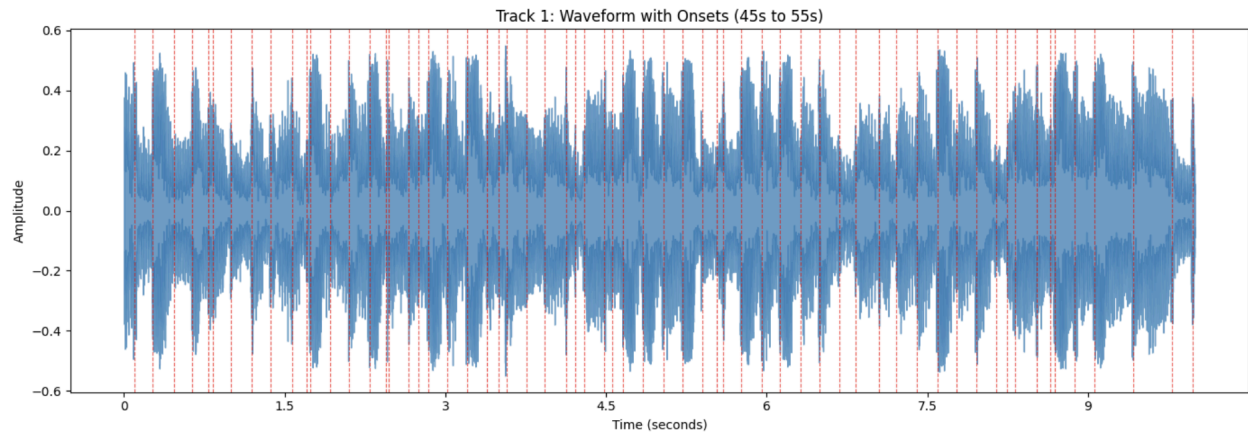


Figure 1: Waveform of Track 1

Track 2: Chopin Nocturne in E-flat Major, Op. 9, No. 2

Tempo Estimation

Librosa estimated the tempo at 126.05 BPM. The actual tempo of this piece is approximately 60 to 70 BPM (Andante), although it fluctuates significantly due to rubato for expressive movement, a performance practice where the pianist deliberately speeds up and slows down for expressive effect. The estimate of 126 BPM suggests the algorithm is again detecting a faster subdivision level rather than the main beat, likely responding to the continuous left-hand arpeggiated accompaniment pattern rather than the broader melodic pulse. Unlike the Tyson Yoshi track, where the octave error is a clean 2:1 ratio, the Chopin estimate does not divide neatly into the actual tempo, which reflects the additional confusion caused by irregular timing.

Onset Detection (30 onsets in 10 seconds, segment from 112s to 122s)

The onset detection identified 30 events, roughly half the number found in the pop track despite similar segment length. This lower count seems reasonable as a solo piano performance has fewer simultaneous sound sources than a full pop mix. However, examining the waveform

plot reveals that onset placement is less reliable here. In legato passages where notes connect smoothly under the sustain pedal, note boundaries are blurred, and the algorithm either fails to recognize some onsets or places them slightly off from the actual note attacks. The pedaled sections create a continuous wash of sound where individual note beginnings are acoustically ambiguous. Additionally, grace notes and ornamental turns characteristic of Chopin's style may generate extra onset detections that may not correspond to structurally significant notes.

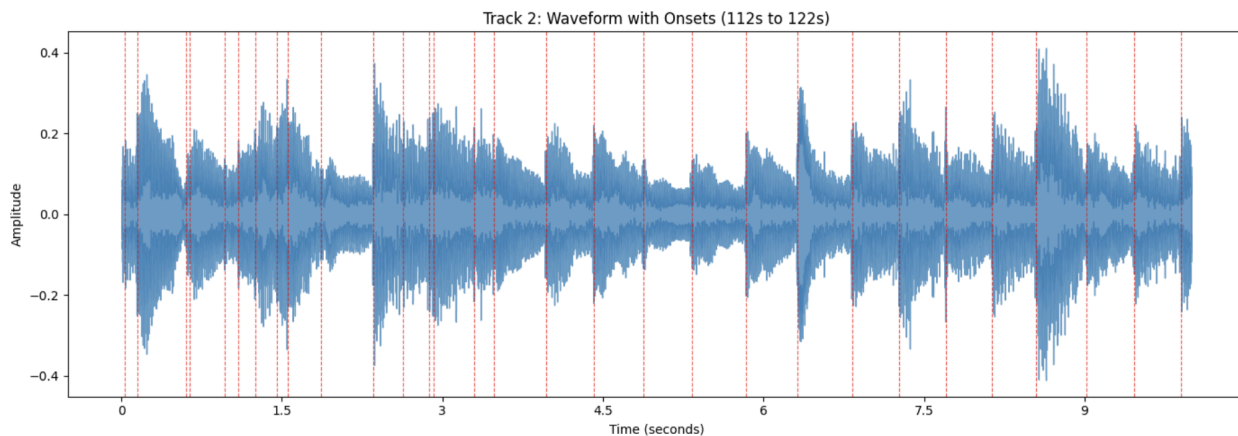


Figure 2: Waveform of Track 2

Comparison and Observations

The two tracks demonstrate a clear pattern in Librosa's performance. The algorithm handles rhythmically regular, percussive music well: the Tyson Yoshi track has a consistent drum machine pattern that provides strong, unambiguous transients for both tempo and onset detection. The main error was the octave doubling of the tempo, which is a known and predictable limitation.

The Chopin piece exposes the algorithm's weaknesses with expressively timed classical music. Rubato breaks the assumption of a steady tempo, the sustain pedal obscures note

boundaries, and the absence of percussion removes the clear transient cues that onset detection relies on. These results suggest that Librosa's beat tracking and onset detection tools are primarily tuned for popular music with steady tempos and strong percussive elements, and they struggle when those assumptions are not met.